

ENEL700 Tutorial 9

1. An information source is characterized by the following alphabet and probability distribution

Alphabet	A1	A2	A3	A4	A5
Probability	0.1	0.4	0.3	0.15	0.05

Compute the Entropy for the source.

$$\begin{aligned}
 >> H(X) &= \sum_{i=1}^m P(x_i) I(x_i) \\
 &= 0.1 \times \log_2(1/0.1) + 0.4 \times \log_2(1/0.4) + 0.3 \times \log_2(1/0.3) \\
 &\quad + 0.15 \times \log_2(1/0.15) + 0.05 \times \log_2(1/0.05) \\
 &= 2.00869 \text{ bits}
 \end{aligned}$$

2. Suppose the following code is used for the information source described in question 2, determine its coding efficiency.

Alphabet	A1	A2	A3	A4	A5
Code	01	0	1	00	001

>> average code length

$$L = \sum_{i=1}^m P(x_i) n_i = 0.1 \times 2 + 0.4 \times 1 + 0.3 \times 1 + 0.15 \times 2 + 0.05 \times 3 = 1.35$$

Therefore the code efficiency is $\eta = H(X)/L = 2.00869/1.35 > 100\%$ which implies that it is impossible to implement this in practice. In fact, the optimal coding by Huffman coding is here, which has $L=2.05$ and $\eta=97.98\%$.

Alphabet	A1	A2	A3	A4	A5
Code	0000	1	01	001	0001

3. The American Standard Code for Information Interchange (ASCII) has 128 characters, which are binary coded. If a certain computer generated 100,000 characters per second, determine the minimum bandwidth required to transmit this signal (Hint: assume a noiseless channel).

>> the data rate generated by the computer is $R_b = 100000 \times \lceil \log_2 128 \rceil = 700 \text{ kbps}$. For noiseless channel, $C = 2B \log_2 M$, assume binary signalling is used ($M=2$), then the minimum required bandwidth is $B = \frac{1}{2} R_b = 350 \text{ kHz}$

4. A television signal has bandwidth of 4.5MHz, and it is sampled, quantized and binary coded to obtain a PCM signal. Suppose the sampling rate is a rate 20% above the Nyquist rate and the number of quantization levels are 1024. Determine the minimum bandwidth required to transmit this signal.

$$\begin{aligned}
 >> B &= R_b / 2 = f_s \lceil \log_2 N \rceil / 2 \\
 &= 0.5 \times (1 + 0.2) \times 2 \times 4.5 \times 10^6 \times \lceil \log_2 N \rceil = 54 \text{ MHz}
 \end{aligned}$$